Watershed Delineation of Soan River Basin using GIS

Watershed Analysis using DEM-based Hydrological Modeling

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Introduction

- **A watershed is an area of land that drains surface runoff to a common outlet such as a river.**
- **Watershed analysis in GIS uses elevation data to understand how water flows over terrain.**
- **This study demonstrates watershed delineation using DEM and hydrological tools in ArcGIS**

Study Area: Soan River Basin, Pakistan

Major tributary of the Indus River.

It drains the Potohar Plateau, which has complex topography.

The basin is hydrologically important due to:

- Flooding issues
- Soil erosion
- Urban expansion near Islamabad & Rawalpindi

Availability of freely accessible DEM data makes it suitable for academic analysis.

Study Area



Objectives of the Study

- To delineate the watershed boundary of the Soan River Basin.
- To generate drainage patterns using DEM-based analysis.
- To understand surface water flow behavior.
- To demonstrate practical use of GIS hydrology tools.

Data Used

Digital Elevation Model (DEM):

- **Source: USGS / SRTM**
- **Resolution: 30 m**
- **Software: ArcGIS Desktop 10.5**

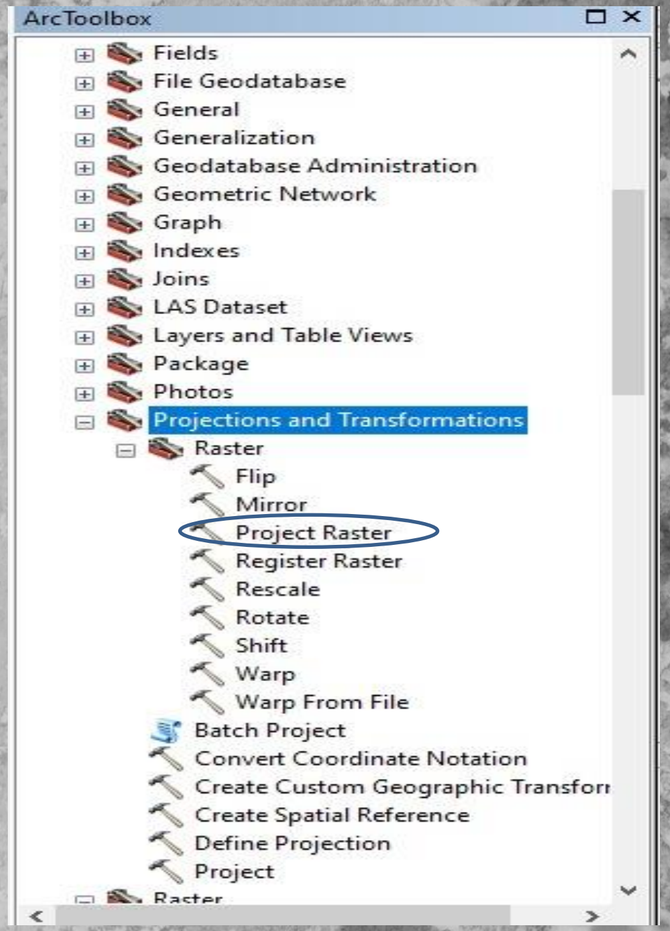
Coordinate System: Geographic Coordinate System (WGS 84)

Methodology Overview

1. DEM Preparation
2. Sink Filling
3. Flow Direction
4. Flow Accumulation
5. Stream Definition
6. Pour Point Selection
7. Watershed Delineation
8. Polygon Conversion
9. Zonal Statistics

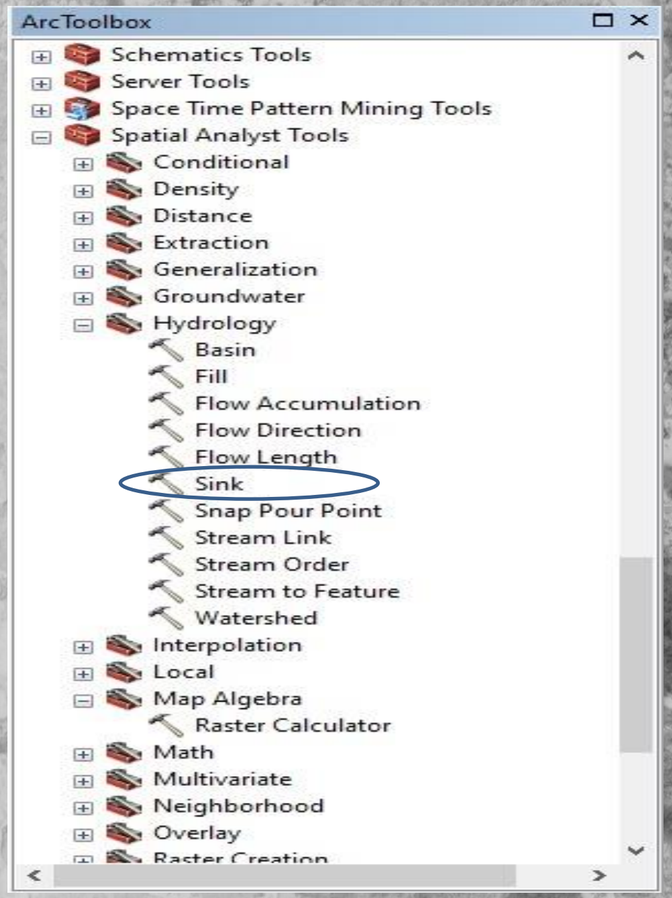
DEM Preparation

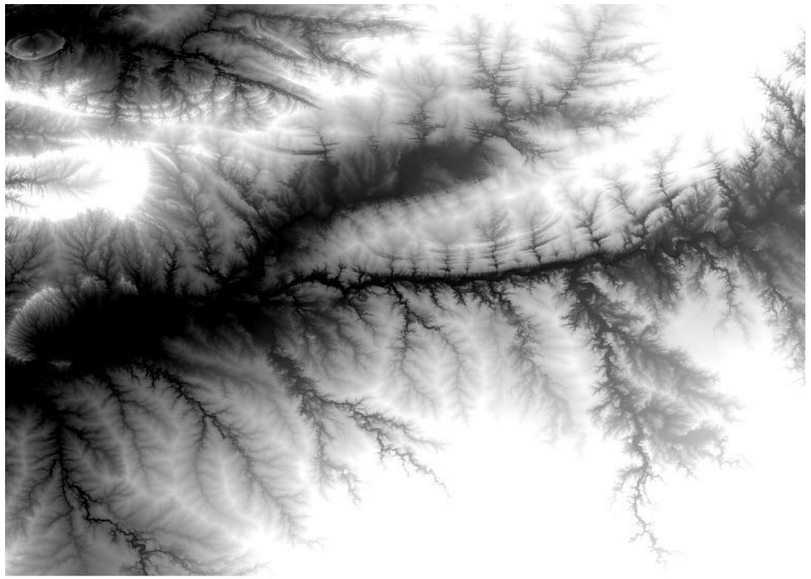
- DEM was clipped to the area of interest.
- DEM was reprojected to PCS coordinates for accurate analysis.
- This DEM represents terrain elevation controlling water flow.



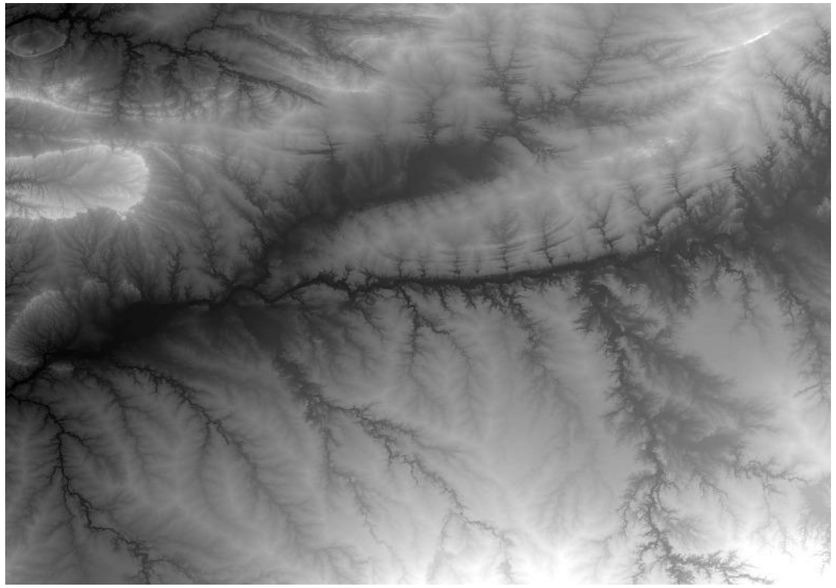
Sink Filling

- DEMs contain small depressions (sinks) that block water flow.
- Sink filling removes these errors.
- This ensures continuous and realistic flow paths.





Raw DEM

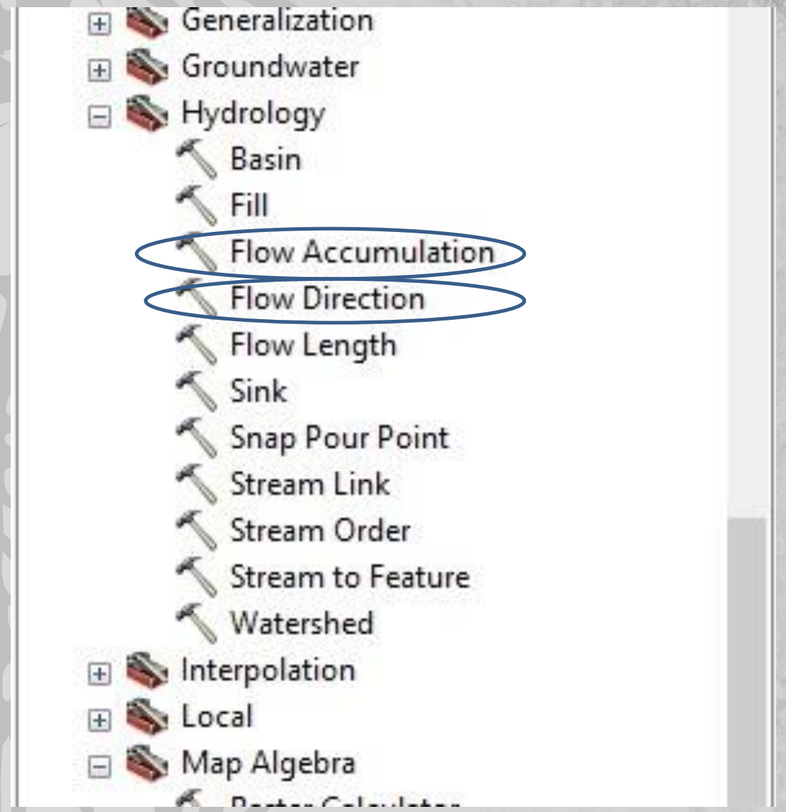


Projected DEM



Filled DEM

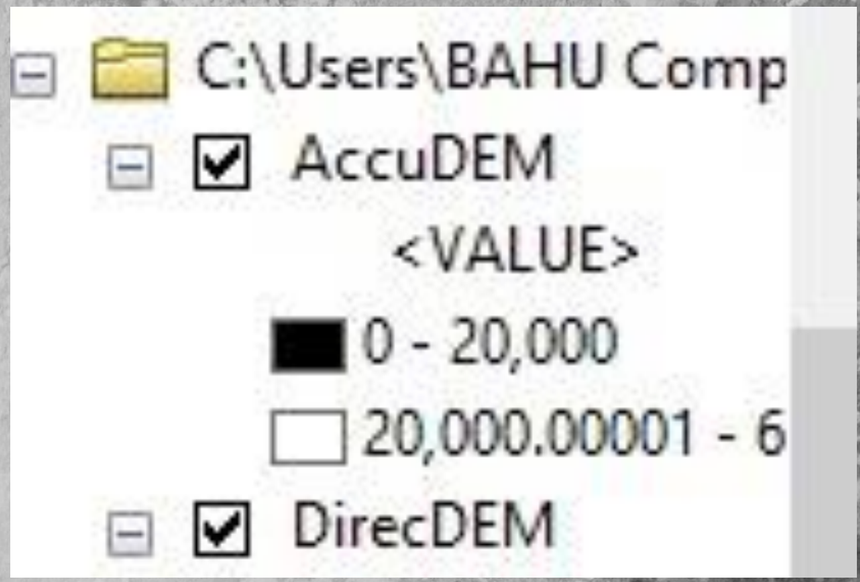
Flow Direction & Flow Accumulation



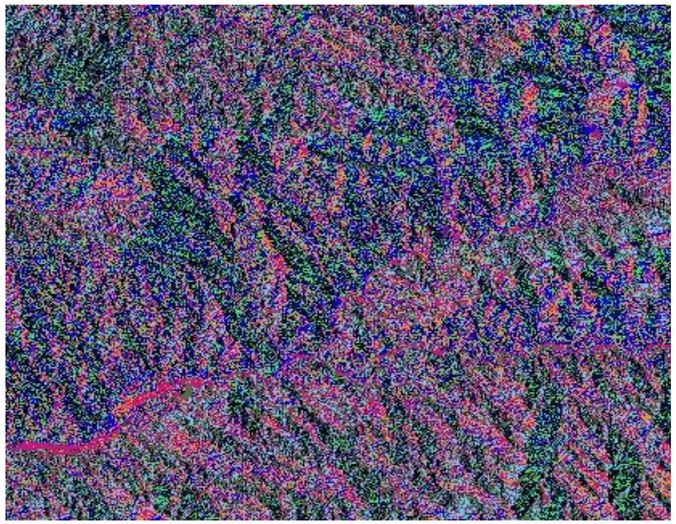
- Flow Direction assigns direction of steepest descent
- Flow Accumulation identifies areas of high runoff
- High values represent potential stream channels

Flow Accumulation

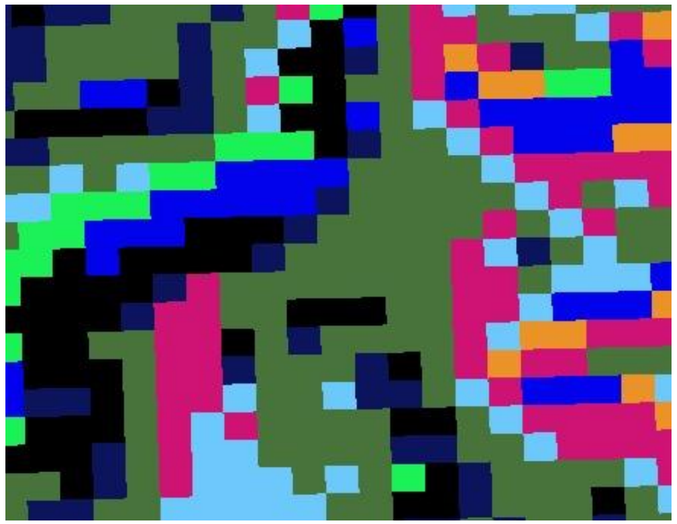
- Flow Accumulation raster generated from Flow Direction.
- Represents number of upstream cells contributing flow.
- Higher values indicate potential stream channels.
- Accumulation threshold modified to 20,000 cells.



Accumulation threshold set to 20,000, so accumulation greater than 20000 will be shown as white streams, rest are ignored



DEM after Flow Direction Tool



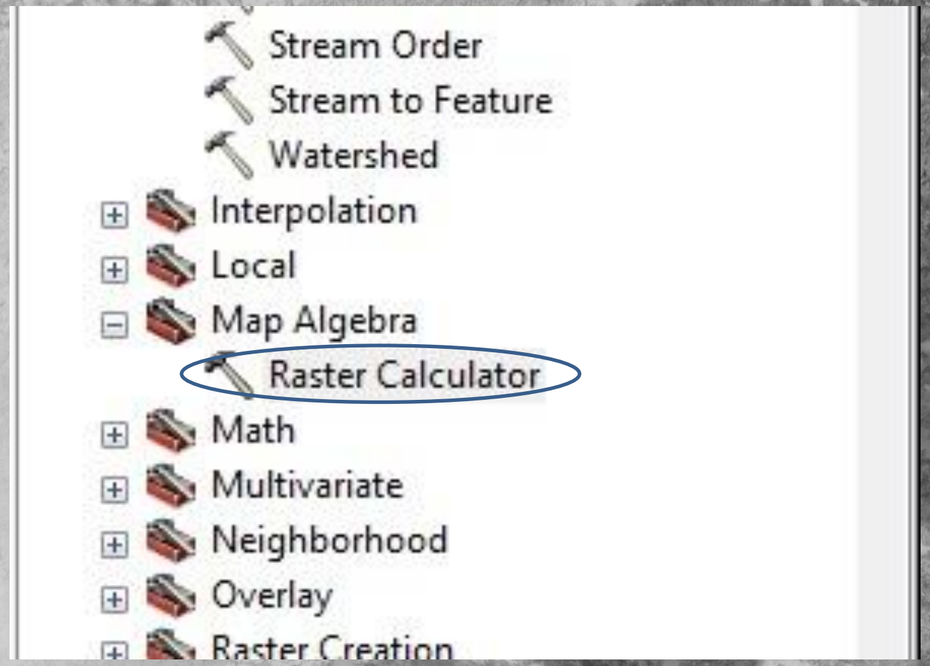
Zoomed DEM after Flow Direction Tool



DEM after Flow Accumulation Tool

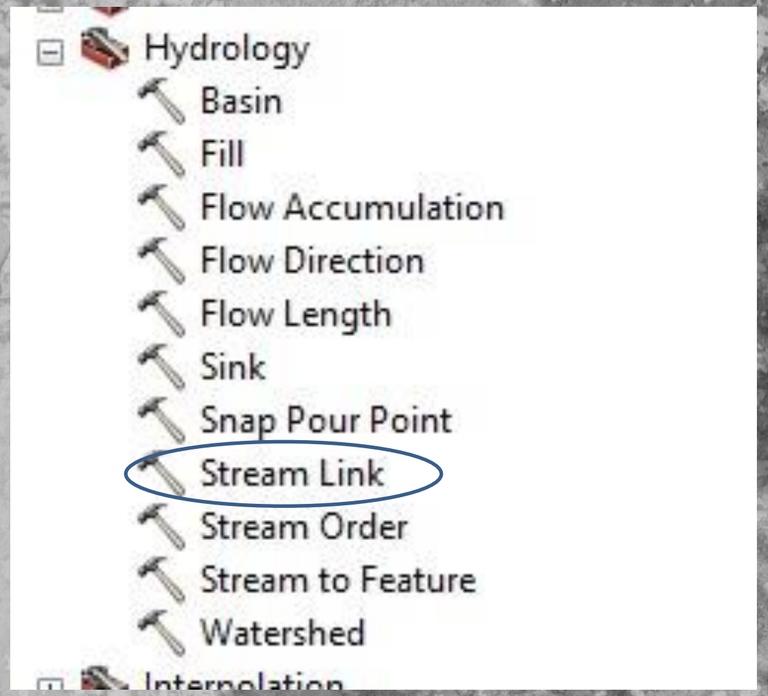
Stream Definition

- Streams extracted using Flow Accumulation threshold.
- Conditional expression applied:
  - Stream = Flow Accumulation > 20,000
- Produces a binary stream raster.
- Represents main drainage network.

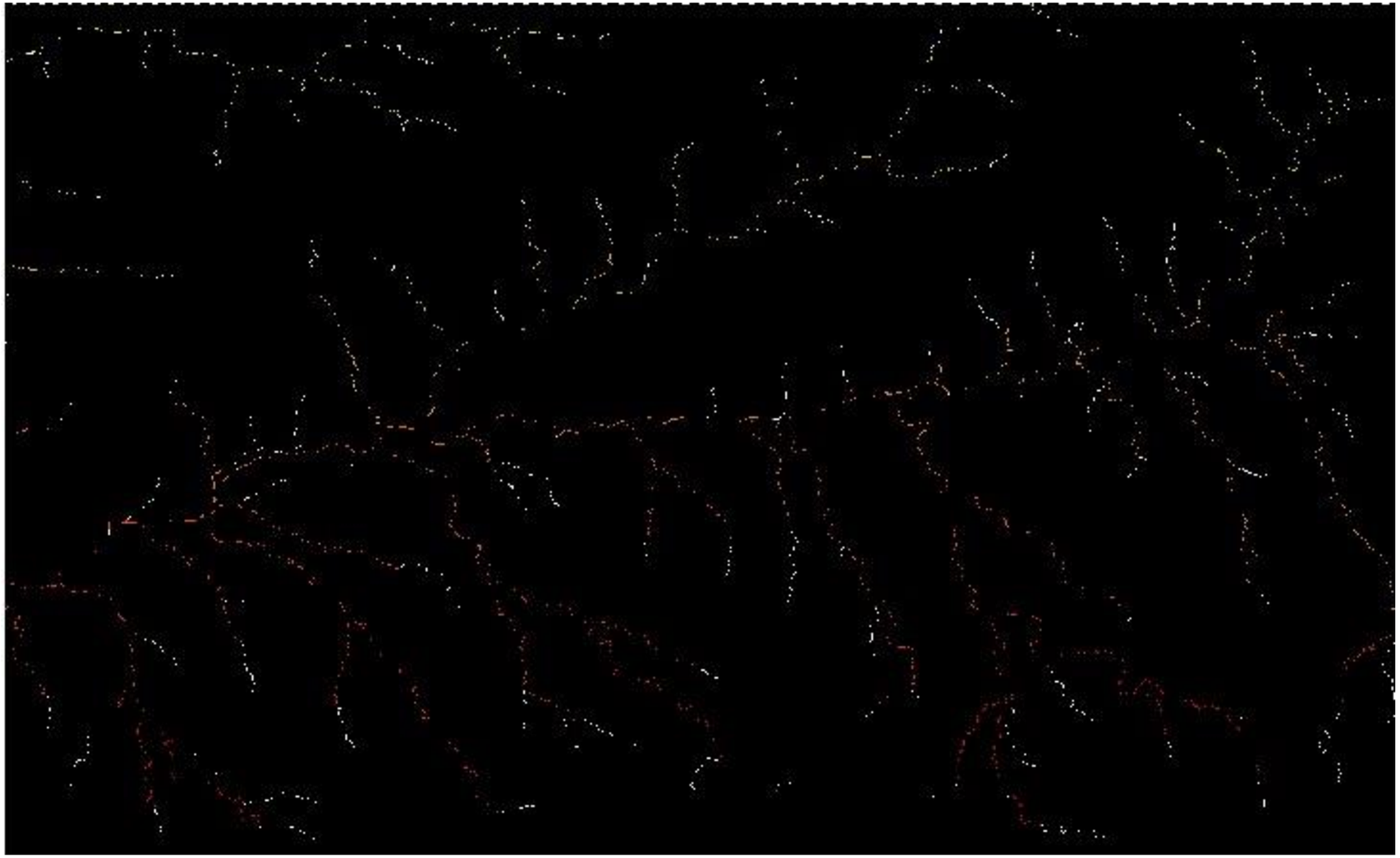


Stream Link

- Stream Link tool applied to stream raster.
- Assigns unique IDs to each stream segment.
- Breaks network at confluences.
- Prepares data for watershed delineation.

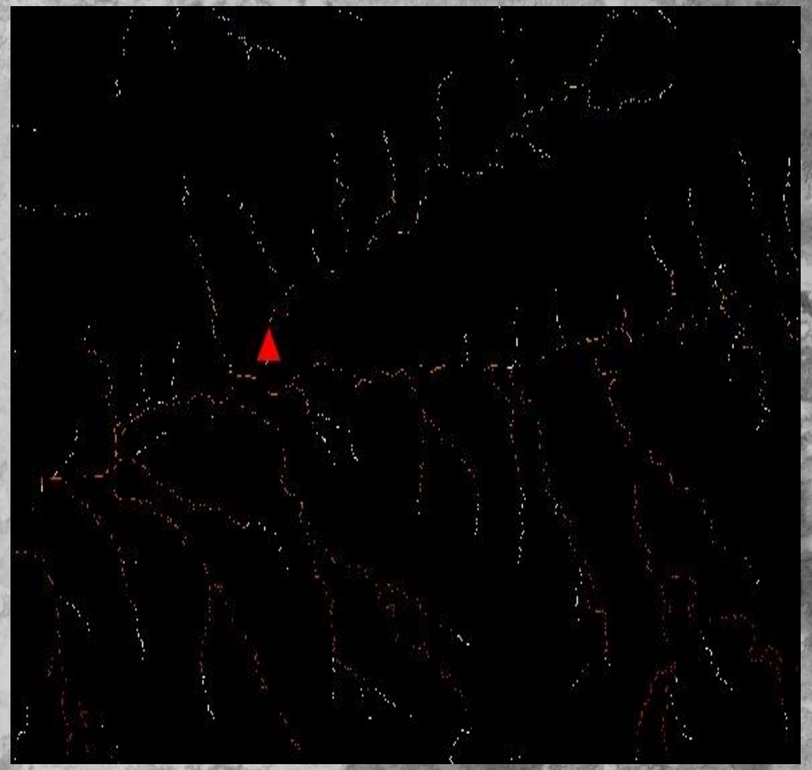


Stream Definition & Stream Link



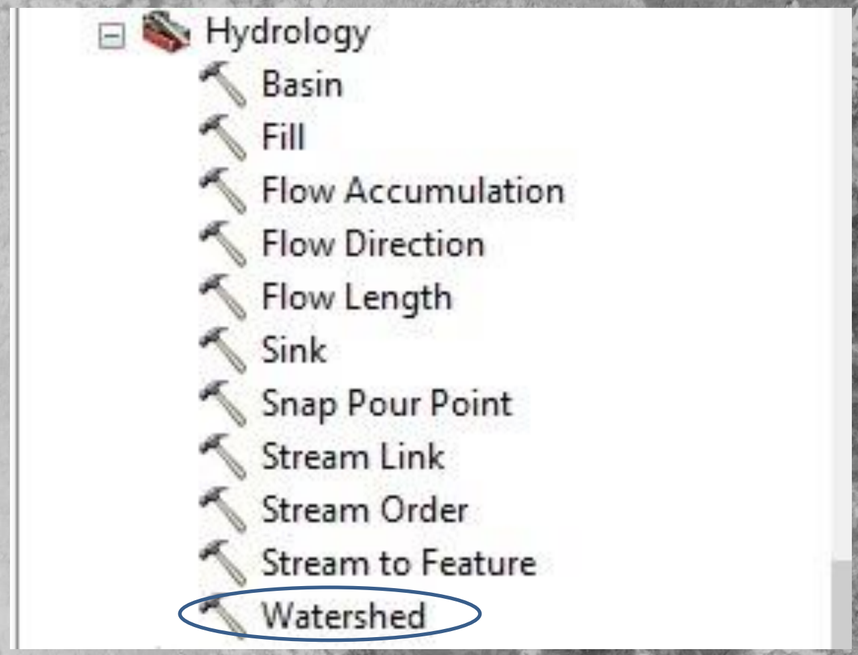
Pour Point Definition

- Placed at **downstream end** of main stream
- Located **directly on stream channel**
- Main stream identified from **stream definition layer**
- **Snap Pour Point** used to align with highest flow accumulation
- Used as **outlet** for watershed delineation

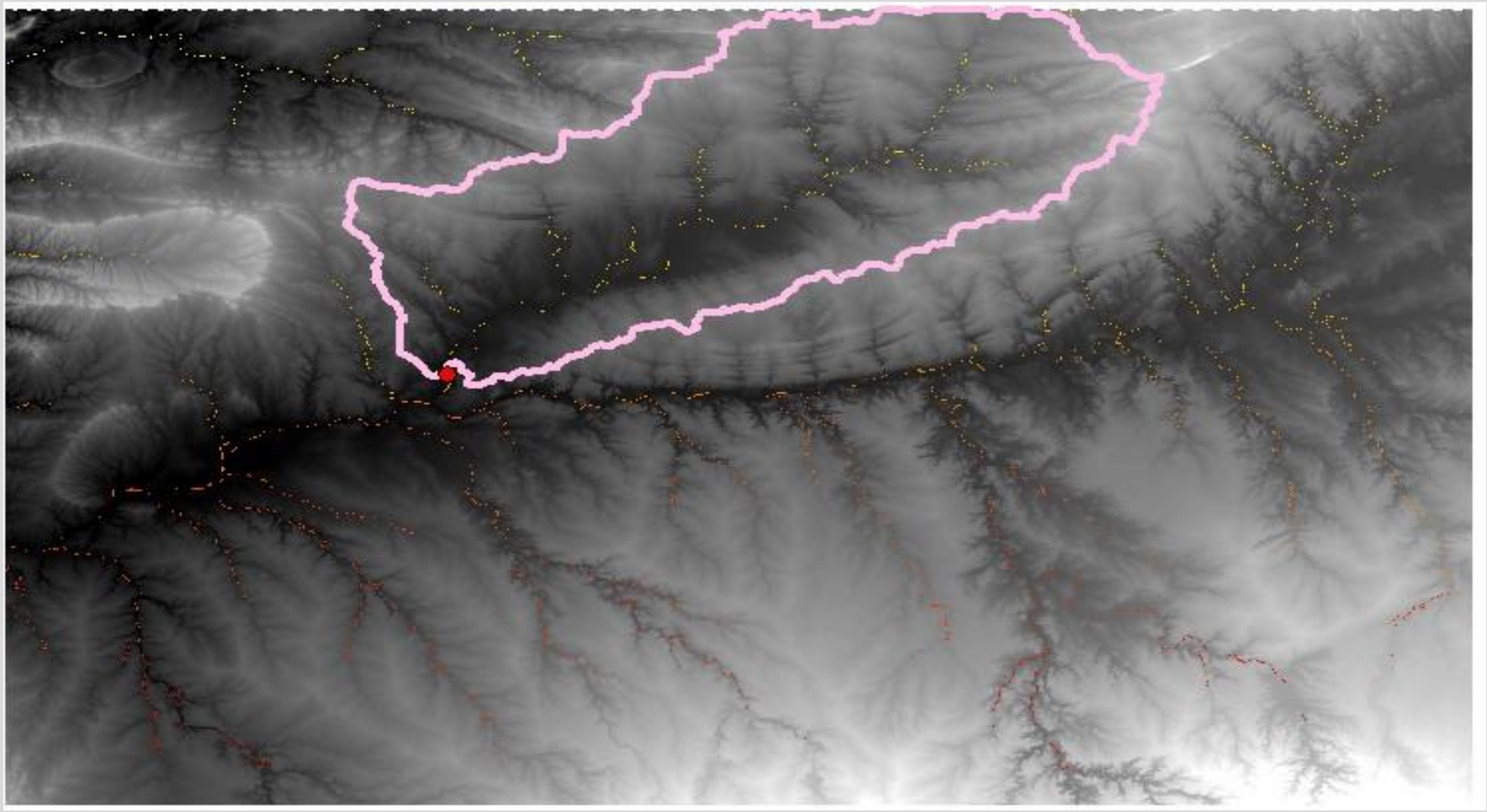


Watershed Delineation

- Watershed delineated using **Flow Direction raster**
- **Snapped pour point** used as outlet input
- Watershed tool traced **all upstream contributing cells**
- Output generated as **watershed raster**
- Raster converted to **watershed polygon** for analysis



Watershed Delineation Output



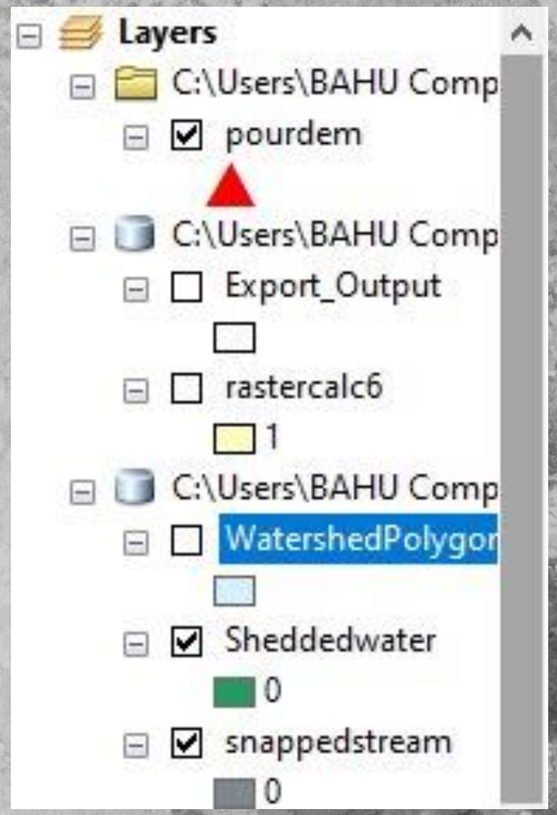
Watershed Polygon Generation

- Watershed raster converted to **vector polygon**
- Conversion done using **Raster to Polygon tool**
- Polygon represents **final watershed boundary**
- Enabled **area calculation and attribute analysis**
- Used for map layout and statistics

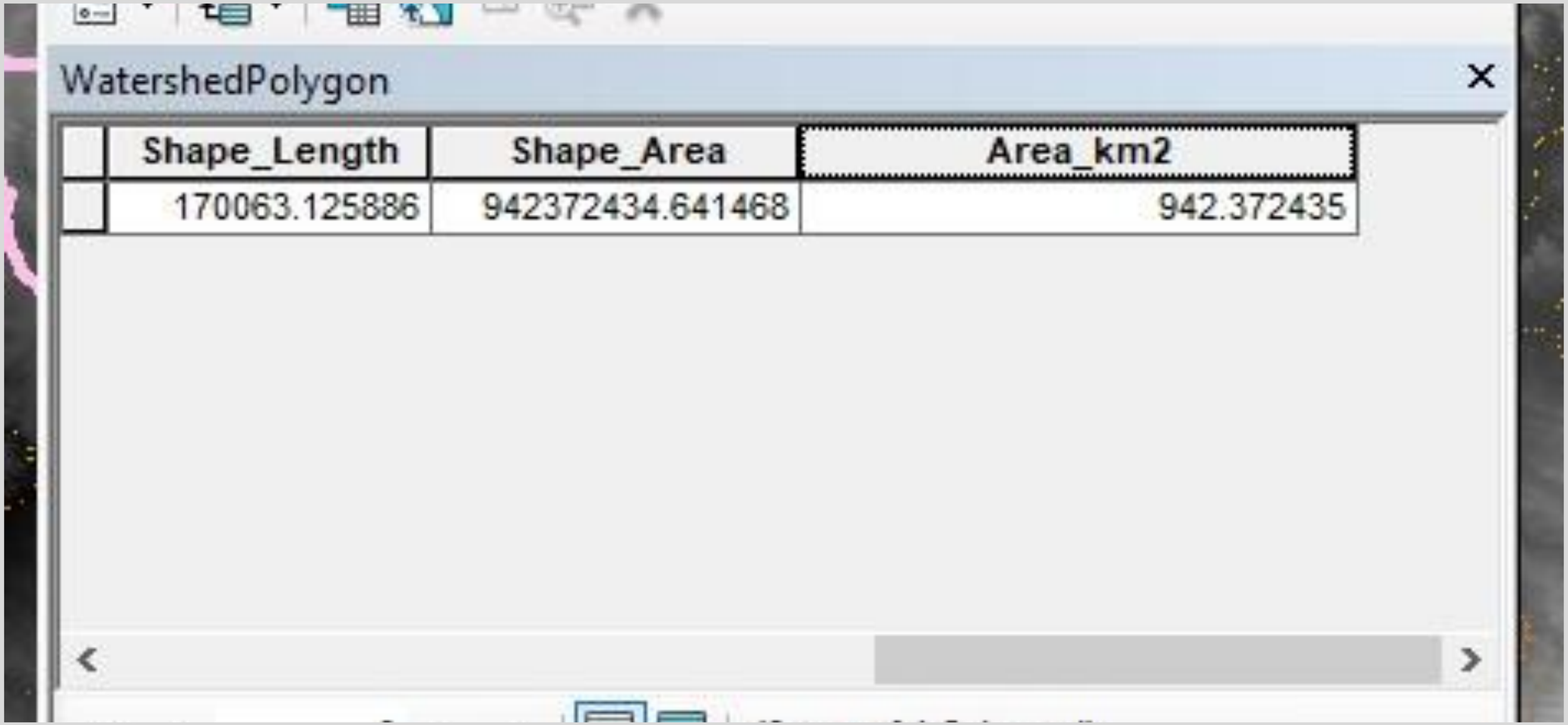


Watershed Area Calculation

- Right click on Watershed Polygon Layer
- Open attribute Table
- Add field of Area
- Add geometry calculation

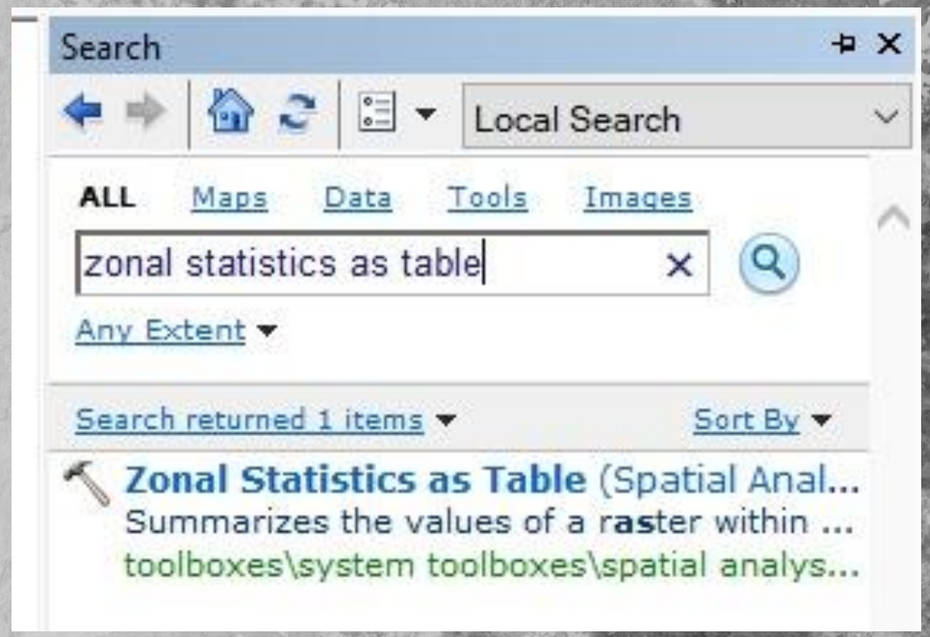


Watershed Area

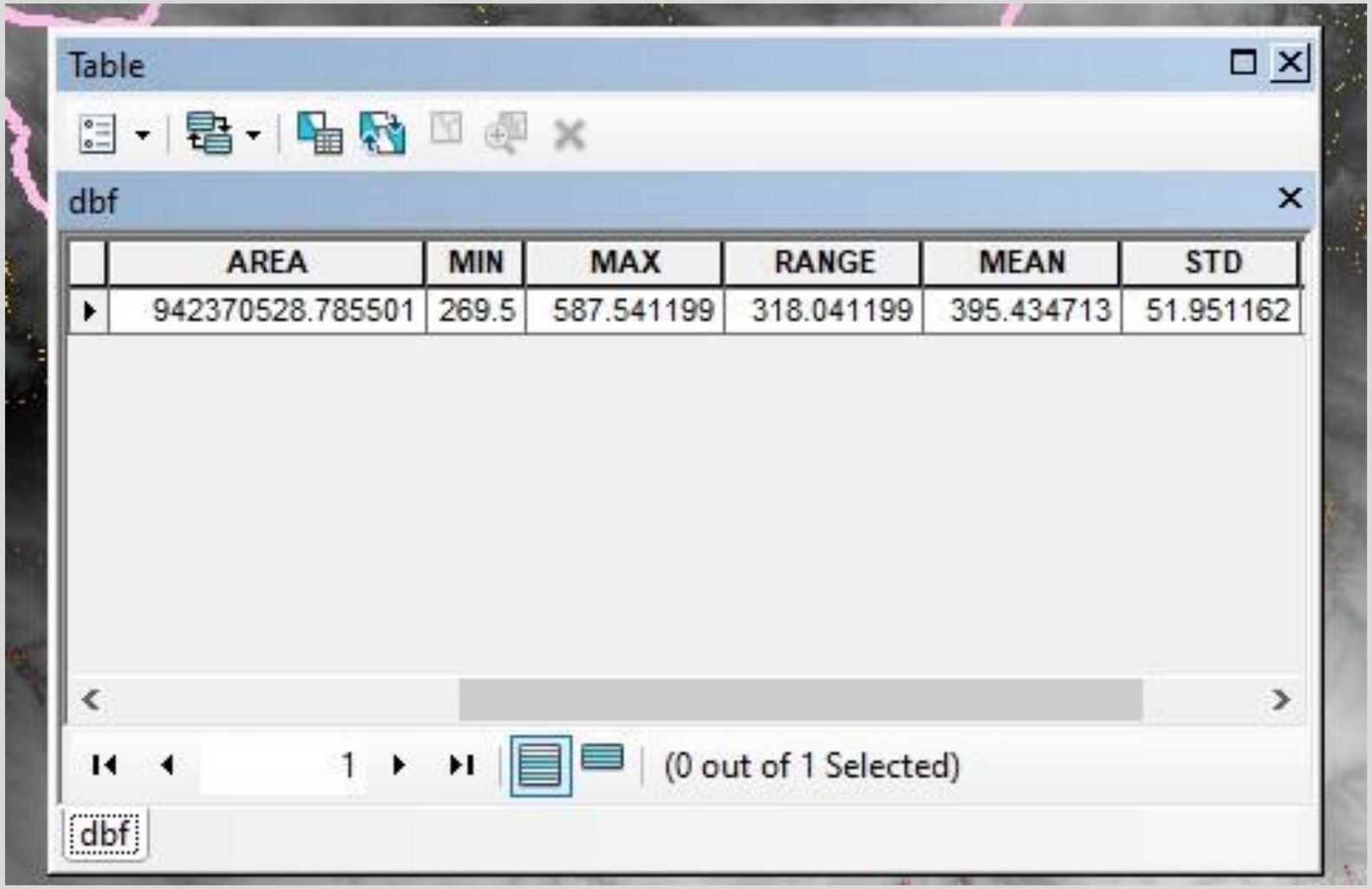


Zonal Statistics

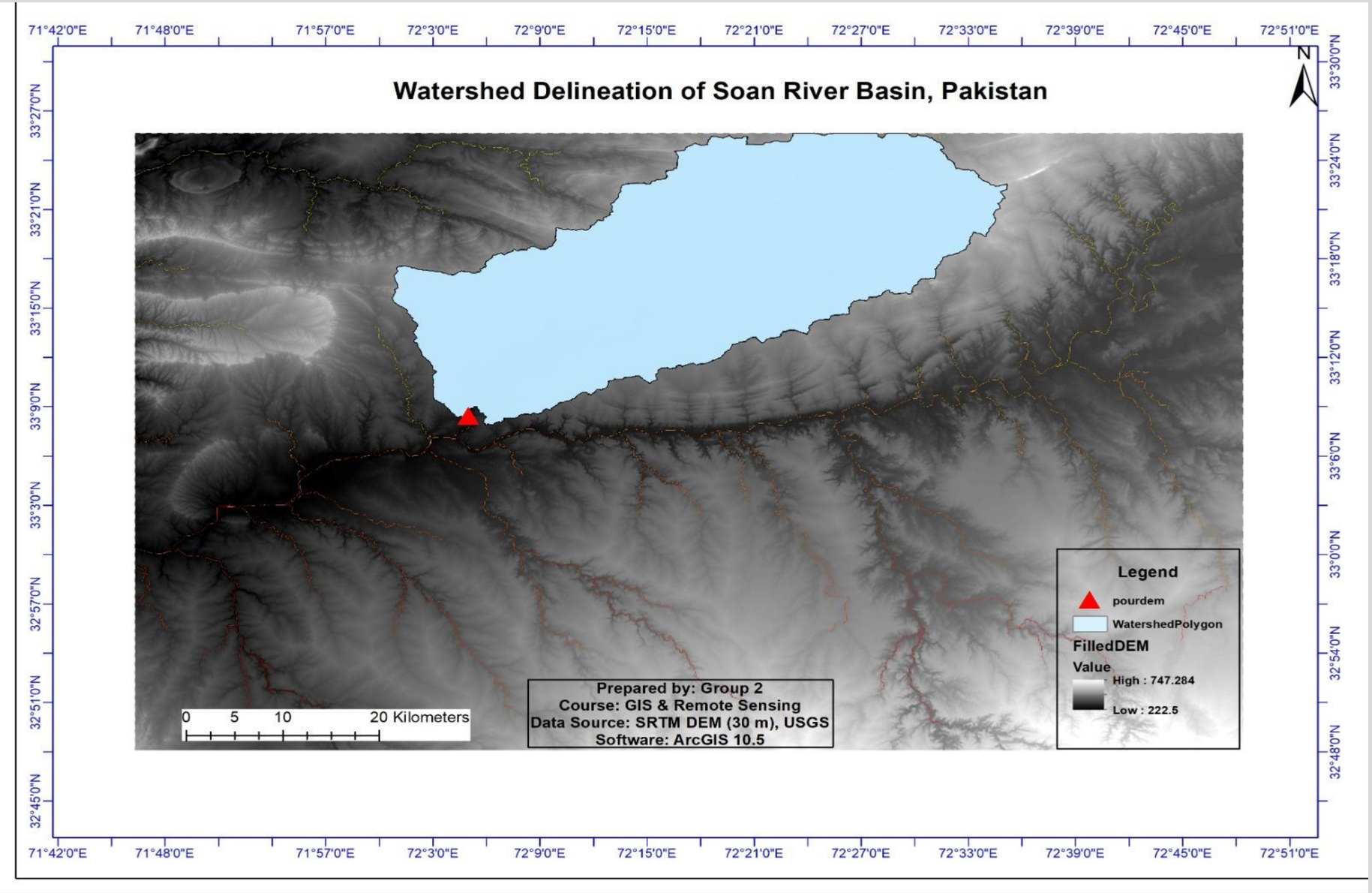
- Zonal Statistics applied using watershed polygon
- DEM used as value raster
- Statistics calculated: min, max, mean elevation
- Output generated as attribute table
- Table joined to watershed polygon



Elevation Characteristics of Watershed



Final Map Layout



Applications

- Flood risk assessment
- Water resource planning
- Soil erosion control
- Dam and reservoir site planning
- Watershed management

Key Learnings

- DEM is the foundation of hydrological modeling
- Pour point accuracy is critical
- Watershed boundaries follow terrain, not administrative limits
- GIS provides quantitative insights for decision-making

Conclusion

- Watershed delineation was successfully performed
- GIS-based hydrological tools are effective and reliable
- The analysis produced accurate spatial and statistical results
- The study demonstrates practical application of GIS in hydrology

Thank You